Physics 1A Discussion (Week 8)

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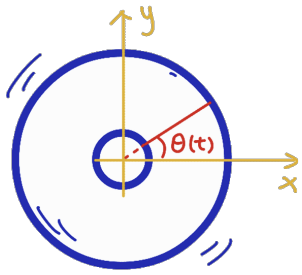


Problem 1: Consider a spinning disk. Jinyan drew a line on the disk, and the line is making a time-dependent angle $\theta(t) = a + bt - ct^3$ with respect to the x-axis as shown, where a, b, and c are constants, and t is measured in seconds. When $t = 0$ seconds, it is found that $\theta = \pi/4$ and the angular velocity of the disk is 2.00 rad/s. When $t = 1.50$ seconds, it is found that the angular acceleration of the disk is 1.25 rad/s^2.

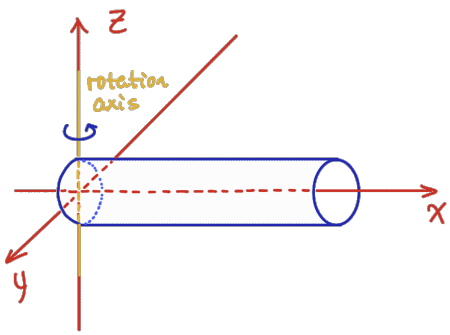
(a) Find a, b, and c based on the given information, including their units.

(b) Find the angular acceleration of the disk when $\theta = \pi/4$ rad.

Problem 2: Show that (by computing the integral) the moment of inertia of a solid rod of length L, radius R, and mass M, rotating about an axis through one end of the rod, is

$$I = \int |\vec{r}_\perp|^2 \, dm = \frac{ML^2}{3} + \frac{MR^2}{4} \, .$$

Assume that the rod has uniform mass density. Here $\vec{r}_\perp$ is a vector perpendicular to the axis of rotation, extending from a point on the rotation axis to a point $\vec{r}$ in the solid rod, and the integral is over the entire rod.



Problem 1: Part (a) is an initial value problem (or boundary value problem when talking about differential equations in general). Here the angular velocity is defined as taking the first derivative of θ with respect to time, denoted as $\dot{\theta}$. The angular acceleration is defined as taking the second derivative of θ with respect to time, denoted as $\ddot{\theta}$.

To find a, b, and c, we need to utilize the information about the disk at $t = 0$ and $t = 2$ given in the problem. There are exactly three data points given, and we need to use all three of them to solve for the values of the three variables a, b, and c. Namely: the first one is $\theta(t = 0) = \pi/4$, which gives

$$\frac{\pi}{4} = \theta(t = 0) = a\,;$$

the second one is $\dot{\theta}(t = 0) = 2$, which gives

$$2 = \frac{d\theta}{dt}|_{t=0} = b - 3ct^2|_{t=0} = b\,;$$

and the last one is $\ddot{\theta}(t = 1.5) = 1.25$, which gives

$$1.25 = \frac{d^2\theta}{dt^2}|_{t=1.5} = -6ct|_{t=1.5}\,,$$

and thus $c = -0.139$. Now for the unit of each quantity, we know that θ is measured in rad, so a should also be measured in rad because a and θ should have the same physical dimension. On the other hand, bt and θ have the same physical dimension, so b is measured in rad per second, same as angular velocity. Lastly, θ and ct^3 have the same physical dimension, so c is measured in rad per second cubed (rad/s^3).

Part (b) is finding the angular acceleration of the disk when $\theta = \pi/4$, first we want to find the time when the disk gets to $\theta = \pi/4$, so we set

$$\frac{\pi}{4} = \frac{\pi}{4} + 2t + 0.139t^3\,,$$

from which we get

$$t = 0\,\text{seconds} \qquad \text{or} \qquad t = \sqrt{-\frac{2}{0.139}}\,\text{seconds}\,.$$

Time cannot be imaginary, so we are only interested in the $t = 0$ seconds case. At time $t = 0$, the acceleration of the disk is

$$\frac{d^2\theta}{dt^2}|_{t=0} = -6ct|_{t=0} = 0\,\text{rad/s}^2\,.$$

Problem 2: We want to evaluate the following integral:

$$I = \int_{\text{rod}} r_\perp^2\, dm = \int\int\int_{\text{rod}} r_\perp^2 \rho\, dx\, dy\, dz\,,$$

where I have used the change of variable

$$dm = \frac{dm}{dV}\, dV = \frac{dm}{dV}\, dx\, dy\, dz = \rho\, dx\, dy\, dz\,.$$

It is mentioned in the problem that the mass density ρ is uniform, that is ρ is a constant, independent on the spatial position, and we can write

$$\rho = \frac{M}{\text{volume of the rod}} = \frac{M}{\pi R^2 L}\,.$$

Anyway, ρ is just a constant, and we can pull it out of the integral, thus the integral becomes

$$I = \rho \int\int\int_{\text{rod}} r_\perp^2 \, dx\, dy\, dz\,.$$

Now we want to figure out what $\vec{r}_\perp$ is. By definition, it is a vector measured from the rotation axis (the z-axis in this case) to a point in the rod, being perpendicular to the rotation axis. Thus, if we have a point $\vec{r} = (x, y, z)$ in the rod, then the corresponding $\vec{r}_\perp$ is the component of $\vec{r}$ that is perpendicular to the rotation axis, so

$$\vec{r}_\perp = (x, y, 0)\,, \qquad |\vec{r}_\perp|^2 = x^2 + y^2\,.$$

Now

$$I = \rho \int\int\int_{\text{rod}} (x^2 + y^2)\, dx\, dy\, dz = \rho \int_{-R}^{R} \int_{-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} \int_0^L (x^2 + y^2)\, dx\, dy\, dz\,.$$

A direct computation is given here,

$$\begin{aligned}
I &= \rho \int_{-R}^{R} \int_{-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} \left[\frac{x^3}{3} + y^2 x\right]_{x=0}^{x=L} dy\, dz \\
&= \rho \int_{-R}^{R} \int_{-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} \left(\frac{L^3}{3} + y^2 L\right) dy\, dz \\
&= \rho \int_{-R}^{R} \left[\frac{L^3 y}{3} + \frac{y^3 L}{3}\right]_{y=-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} dz \\
&= \rho \int_{-R}^{R} \left(\frac{2L^3}{3}\left(R^2 - z^2\right)^{1/2} + \frac{2L}{3}\left(R^2 - z^2\right)^{3/2}\right) dz \\
&= \rho \left(\frac{\pi L^3 R^2}{3} + \frac{\pi L R^4}{4}\right) \qquad (*) \\
&= \frac{ML^2}{3} + \frac{MR^2}{4}\,.
\end{aligned}$$

Here I used online integral calculator to brute force calculating the dz integral because I was lazy. It is also possible to carry out the dz integral with a change of variable

$$z = R\sin(\theta)\,, \qquad \text{s.t. } dz = R\cos(\theta)\, d\theta\,.$$

However, we can also use symmetry to do the $dy\, dz$ double integral. In this case, we are integrating over a circular disk on the yz-plane,

$$\int\int_{\text{disk}} \left(\frac{L^3}{3} + y^2 L\right) dy\, dz = \frac{L^3}{3} \int\int_{disk} dy\, dz + L \int\int_{\text{disk}} y^2\, dy\, dz\,.$$

The first term is just

$$\frac{L^3}{3} \cdot (\text{area of the disk}) = \frac{\pi L^3 R^2}{3}$$

which is exactly the first term we obtained in (*). The second term in the disk integral can be evaluated using the symmetry

$$\int\int_{\text{disk}} y^2\, dy\, dz = \int\int_{\text{disk}} z^2\, dy\, dz\,.$$

Here $y^2 + z^2 = r^2$ where r is the radial polar coordinate on the disk, thus we use polar coordinate integration to compute

$$\int\int_{\text{disk}} r^2\, r\, dr\, d\theta = \int_0^{2\pi}\int_0^R r^3\, dr\, d\theta = \frac{\pi R^4}{2}\,.$$

Thus we see

$$L\int\int_{\text{disk}} y^2\, dy\, dz = \frac{L}{2}\int\int_{\text{disk}} (y^2+z^2)\, dy\, dz = \frac{L}{2}\int\int_{\text{disk}} r^2\, r\, dr\, d\theta = \frac{\pi L R^4}{4}\,,$$

which is exactly the second term we expect to see in the previous calculation from (*).

Now, what if the radius of the base of the cylinder is approaching zero, that is $R \sim 0$, from the calculation above, we expect to see

$$I = \frac{ML^2}{3}\,.$$

Let us verify this: This setup is sometimes referred to the "thin rod," where the radius of the rod is negligible and we can essentially treat the rod as a one-dimensional object (like a string), so the corresponding integral for computing I will be a one-dimensional integral, in the x-direction:

$$I = \int_{\text{rod}} r_\perp^2\, \rho\, dx\,.$$

The lower bound of the integral is the left end of the rod at $x = 0$, and the upper bound is the right end at $x = L$. Here $r_\perp = x$ because the rod is thin enough that we can treat $y = 0$. Therefore,

$$I = \int_0^L x^2\, \rho\, dx = \frac{\rho L^3}{3}\,.$$

The mass density ρ becomes 1-dimensional because the thin rod is 1-dimensional, so $\rho = M/L$. Finally,

$$I = \frac{ML^2}{3}\,.$$

This calculation is much simpler, and is something that we expect you to be able to do during quizzes or exams.